

Decision Memo — Retail Demand Forecasting & Replenishment Planner Capstone

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**Retail Demand Forecasting &
& Replenishment
Planner**



Retail Demand Forecasting & Inventory Replenishment Planner

This memo summarizes the forecasting, inventory risk monitoring, and replenishment planning workflow created for store-SKU level retail operations. The goal is to improve fill rate, reduce stockouts, and support more data-backed purchase planning.

1. Goal and Key Performance Indicators (KPIs)

Retail businesses often face challenges related to inventory imbalance, which includes both stockouts and overstock situations. Stockouts result in lost sales and dissatisfied customers, while overstock increases inventory holding costs and reduces operational efficiency.

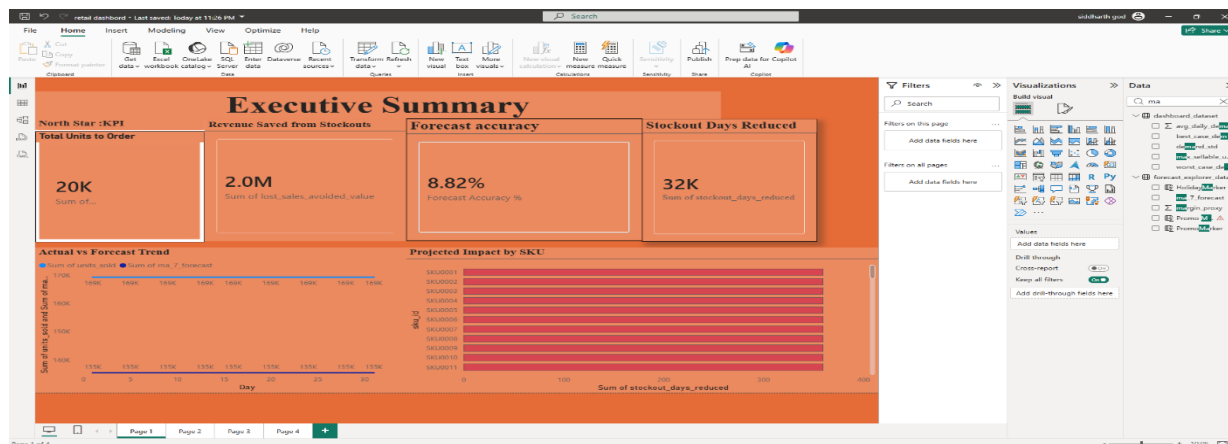
The goal of this project is to develop a data-driven demand forecasting and inventory replenishment planning system that helps retailers anticipate product demand and optimize inventory decisions.

The system integrates forecasting analytics with inventory planning to support operational decision making.

The key performance indicators used to measure the system's performance include:

- Total Units to Order – The number of units that need to be ordered to maintain adequate inventory levels.
- Revenue Saved from Stockouts – Estimated revenue preserved by preventing product shortages.
- Forecast Accuracy – The accuracy of predicted demand compared with actual sales.
- Stockout Days Reduced – Estimated number of stockout days prevented through proactive planning.

These KPIs provide managers with a quick overview of inventory performance and highlight areas requiring intervention.



1.Data and Methodology

The analysis uses retail sales and inventory data at the **SKU and store level**.

Key variables included in the dataset are:

- SKU identifier
- Store identifier
- Units sold
- Forecast demand
- Inventory on hand
- Promotion indicators
- Holiday indicators
- Date information

To prepare the data for analysis, an **ETL (Extract, Transform, Load) process** was performed

Data Cleaning

The dataset was cleaned to remove inconsistencies and missing values. SKU and store identifiers were standardized to ensure accurate aggregation and analysis.

Feature Engineering

Several new variables were created to improve forecasting and inventory analysis:

- Daily average demand
- Promotion markers
- Holiday markers
- Lagged sales variables for forecasting

These features help capture demand patterns and identify external factors that influence product sales.

Data Aggregation

Sales data was aggregated by date, SKU, and store to enable time-series analysis and dashboard visualization. A date dimension table was also created to support trend analysis and filtering.

3. Forecast Performance and Forecasting Approach

Demand forecasting is an essential component of supply chain planning. For this project, a 7-day moving average forecasting model was used.

The moving average model predicts future demand by calculating the average of sales from the previous seven days. This method is widely used in retail operations because it provides stable short-term forecasts and is easy to interpret.

Forecast accuracy is calculated using the following formula:

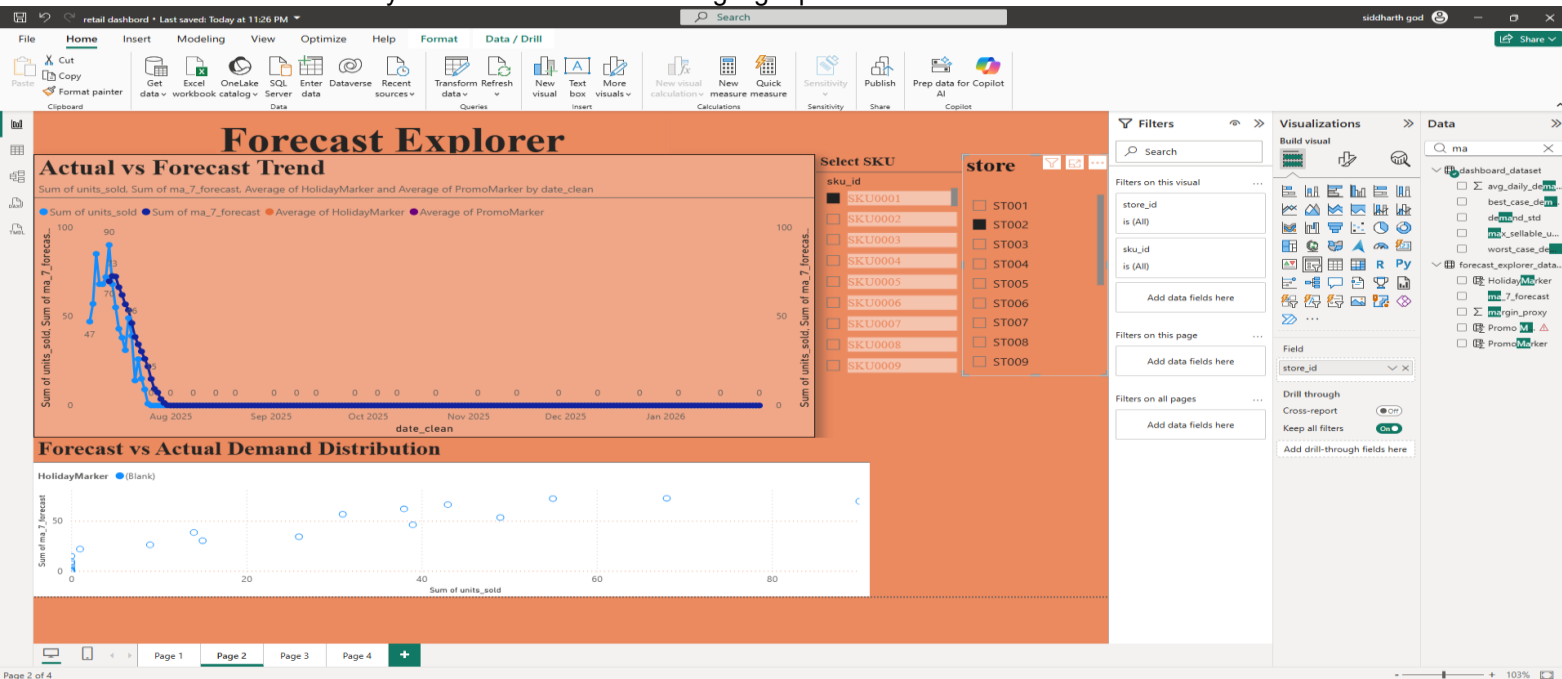
$$\text{Forecast Accuracy} = 1 - (\text{Absolute Forecast Error} \div \text{Actual Demand})$$

This measure indicates how closely predicted demand aligns with actual sales.

Although the moving average model provides reasonable short-term predictions, forecast accuracy can be affected by sudden demand spikes caused by promotions or seasonal events.

Figure:2 Forecast Explorer Dashboard

The forecast explorer dashboard compares **actual demand with forecasted demand** across different SKUs and stores. Promotion and holiday markers are included to highlight periods where demand fluctuations occur.



4. Key Insights from the Analysis

Several important insights were identified from the dashboard analysis.

High Stockout Risk

Certain SKUs show low inventory coverage and high demand levels, indicating a strong risk of stockouts if replenishment is not performed quickly.

Impact of Promotions

Promotional campaigns significantly increase demand, often causing sudden spikes in sales that can lead to temporary stock shortages.

Forecast Variability Across Products

Demand patterns vary across SKUs, meaning that some products are easier to forecast than others.

Inventory Imbalance Across Stores

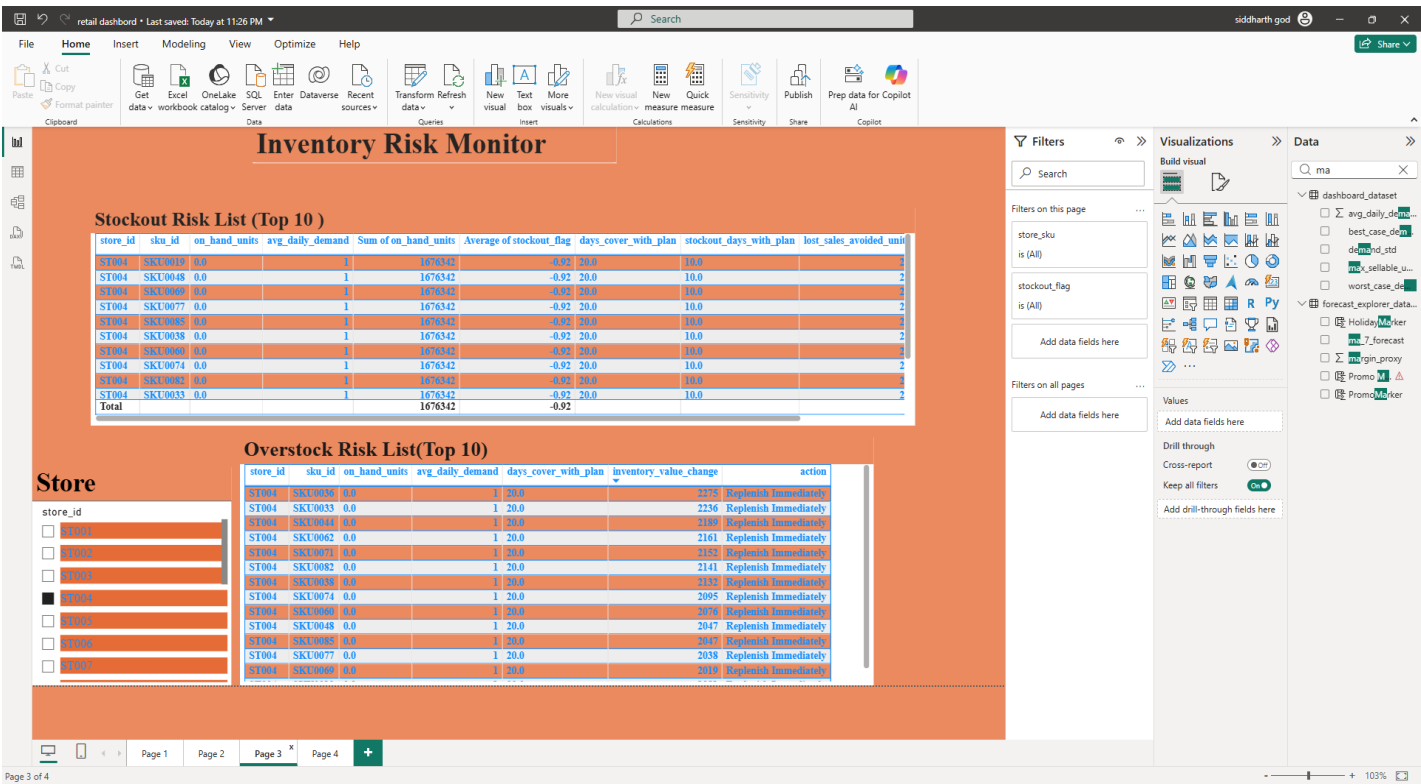
Some stores maintain excess inventory while others experience shortages, suggesting opportunities for better inventory distribution.

Concentration of Risk

A relatively small number of SKUs contribute to the majority of stockout risk. Targeted replenishment strategies can therefore significantly improve inventory performance.

Figure 3: Inventory Risk Monitor Dashboard

The inventory risk monitor identifies SKUs that are most likely to experience stockouts and highlights products with excess inventory.



5. Replenishment Planning Framework

The replenishment planning system calculates how much inventory should be ordered and when orders should be placed.

Three key calculations are used in the replenishment policy.

Safety Stock

Safety stock protects against unexpected demand fluctuations.

Safety Stock =
 $Z \times \text{Demand Standard Deviation} \times \sqrt{\text{Lead Time}}$

Reorder Point

The reorder point determines when a new order should be placed.

Reorder Point =
 $(\text{Average Daily Demand} \times \text{Lead Time}) + \text{Safety Stock}$

Recommended Order Quantity

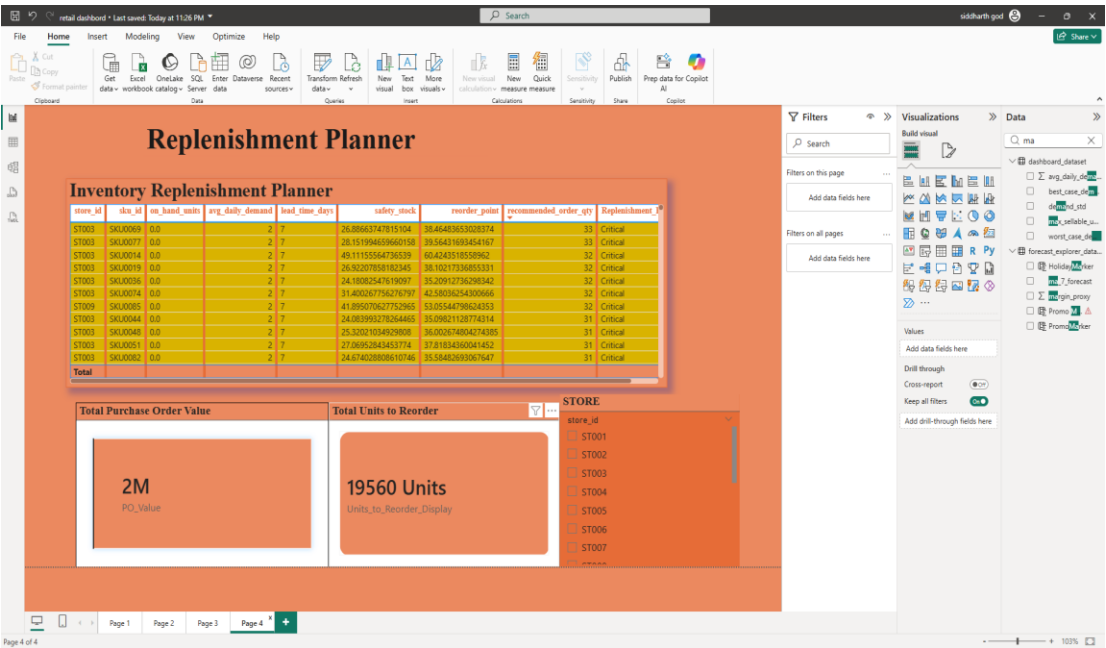
The recommended order quantity is calculated using:

Recommended Order Quantity =
 $\text{Reorder Point} - \text{Current Inventory}$

If the calculated value is negative, no additional inventory needs to be ordered.

Figure 4: Replenishment Planner Dashboard

The replenishment planner dashboard generates recommended purchase orders based on inventory levels, safety stock requirements, and forecast demand.



6 . 30-Day Impact Estimate

To evaluate the potential benefits of the proposed system, a 30-day impact estimate was conducted.

Several assumptions were made:

- Demand patterns remain consistent with recent historical trends
- Supplier lead times remain stable
- No extreme disruptions occur in supply or demand

Based on these assumptions, the replenishment strategy is expected to produce the following benefits:

- Reduced stockout occurrences
- Improved product availability
- Increased revenue due to avoided lost sales
- More balanced inventory distribution across stores

These improvements contribute to better supply chain performance and customer satisfaction.

7. Risks, Limitations, and Next Steps

Although the system provides valuable insights, several limitations must be considered.

Limitations

- The forecasting model does not fully capture seasonal demand patterns.
- Promotions may cause unpredictable demand spikes.
- Supplier lead time variability is not incorporated in the model.
- Forecast accuracy depends on historical data quality.

Future Improvements

Future enhancements could include:

- Implementing advanced forecasting models such as ARIMA or machine learning approaches.
- Incorporating seasonal demand patterns into forecasting.
- Modeling supplier lead-time variability.
- Integrating the dashboard with enterprise resource planning (ERP) systems.
- Automating replenishment order generation.

These improvements would further enhance forecasting accuracy and supply chain decision-making.

